Día Mes Año

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Productos de senos y cosenos

$$\int \operatorname{Sen}^3 2x \cos^4 2x \, dx$$

La potencia del seno es impar: Separamos un factor $\operatorname{Sen} 2x$ y usamos $\operatorname{Sen}^2 2x = 1 - \cos^2 2x$

$$\int \operatorname{Sen}^2 2x \cos^4 2x \operatorname{Sen} 2x \, dx = \int (1 - \cos^2 2x) \cos^4 2x \operatorname{Sen} 2x \, dx$$

Sea $u = \cos 2x \Rightarrow du = -2 \operatorname{Sen} 2x \, dx \Rightarrow \operatorname{Sen} 2x \, dx = -\frac{1}{2} du$

$$= -\frac{1}{2} \int (1 - u^2) u^4 \, du = -\frac{1}{2} \int (u^4 - u^6) \, du = -\frac{1}{2} \left(\frac{u^5}{5} - \frac{u^7}{7} \right) + C$$

$$\int \operatorname{Sen}^3 2x \cos^4 2x \, dx = -\frac{\cos^5 2x}{10} + \frac{\cos^7 2x}{14} + C$$

2. Sustitución trigonométrica

1) $\int \frac{dx}{(4-x^2)^{3/2}}$

Forma $\sqrt{a^2-x^2}$ como $a=2$ $x=2\operatorname{sen}\theta$ $dx=2\cos\theta\,d\theta$

$\sqrt{4-x^2}=2\cos\theta$, $(4-x^2)^{3/2}=8\cos^3\theta$

$$\int \frac{2\cos\theta\,d\theta}{8\cos^3\theta}=\frac{1}{4}\int \sec^2\theta\,d\theta=\frac{1}{4}\tan\theta+C$$

Del triangulo C $\operatorname{sen}\theta=x/2$, cateto adyacente $\sqrt{4-x^2}$:

$$\tan\theta=\frac{x}{\sqrt{4-x^2}}$$

$$\int \frac{dx}{(4-x^2)^{3/2}}=\frac{x}{4\sqrt{4-x^2}}+C$$

2) $\int \frac{\sqrt{25-x^2}}{x}\,dx$

$x=5\operatorname{sen}\theta$, $dx=5\cos\theta\,d\theta$, $\sqrt{25-x^2}=5\cos\theta$

$$\int \frac{5\cos\theta}{5\operatorname{sen}\theta}\,5\cos\theta\,d\theta=5\int \frac{\cos^2\theta}{\operatorname{sen}\theta}\,d\theta=5\int \frac{1-\operatorname{sen}^2\theta}{\operatorname{sen}\theta}$$

$$=5\int(\csc\theta-\operatorname{sen}\theta)\,d\theta$$

$$=5\ln|\csc\theta-\cot\theta|+5\cos\theta+C$$

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Regresando: $\csc\theta = \frac{5}{x}$, $\cot\theta = \frac{\sqrt{25-x^2}}{x}$, $5\cos\theta = \sqrt{25-x^2}$

$$\int \frac{\sqrt{25-x^2}}{x}\,dx = 5\ln\left|\frac{5-\sqrt{25-x^2}}{x}\right| + \sqrt{25-x^2} + C$$

3) $$\int \frac{dx}{x^2\sqrt{a^2-x^2}}$$

$x = a\,\text{Sen}\,\theta$, $dx = a\cos\theta\,d\theta$, $\sqrt{a^2-x^2} = a\cos\theta$

$$\int \frac{a\cos\theta\,d\theta}{a^2\,\text{Sen}^2\theta \cdot a\cos\theta} = \frac{1}{a^2}\int \csc^2\theta\,d\theta = -\frac{1}{a^2}\cot\theta + C$$

$$\cot\theta = \frac{\sqrt{a^2-x^2}}{x}$$

$$\int \frac{dx}{x^2\sqrt{a^2-x^2}} = -\frac{\sqrt{a^2-x^2}}{a^2 x} + C$$

4) $$\int \sqrt{x^2+4}\,dx$$ forma $\sqrt{x^2+a^2}$:

$x = 2\tan\theta$, $dx = 2\sec^2\theta\,d\theta$, $\sqrt{x^2+a^2} = 2\sec\theta$

$$\int 2\sec\theta \cdot 2\sec^2\theta\,d\theta = 4\int \sec^3\theta\,d\theta$$

Integral de $\sec^3\theta$ (por partes): $u = \sec\theta$

$dv = \sec^2\theta\, d\theta \Rightarrow du = \sec\theta\tan\theta\, d\theta$

$v = \tan\theta$

$$\int \sec^3\theta = \sec\theta\tan\theta - \int \sec\theta\tan^2\theta = \sec\theta\tan\theta$$

$$-\int \sec^3\theta + \int \sec\theta$$

$$\Rightarrow 2\int \sec^3\theta = \sec\theta\tan\theta + \ln|\sec\theta + \tan\theta|$$

$$\Rightarrow \int \sec^3\theta = \frac{1}{2}(\sec\theta\tan\theta + \ln|\sec\theta + \tan\theta|)$$

entonces $4\int \sec^3\theta\, d\theta = 2\sec\theta\tan\theta + 2\ln|\sec\theta + \tan\theta| + C$, con $\tan\theta = \frac{x}{2}$, $\sec\theta = \frac{\sqrt{x^2+4}}{2}$

$$= 2\cdot\frac{\sqrt{x^2+4}}{2}\cdot\frac{x}{2} + 2\ln\left|\frac{x+\sqrt{x^2+4}}{2}\right| + C = \frac{x\sqrt{x^2+4}}{2} +$$

$$2\ln(x+\sqrt{x^2+4}) \underbrace{-2\ln 2 + C}_{C_1}$$

$$\int \sqrt{x^2+4}\, dx = \frac{x\sqrt{x^2+4}}{2} + 2\ln(x+\sqrt{x^2+4}) + C$$

5) $\int \frac{x^2\,dx}{(a^2-x^2)^{3/2}}$ $\quad x = a\,\operatorname{Sen}\theta,\ dx = a\cos\theta\,d\theta,$

$(a^2-x^2)^{3/2} = a^3\cos^3\theta$

$$\int \frac{a^2\operatorname{sen}^2\theta \cdot a\cos\theta\,d\theta}{a^3\cos^3\theta} = \int \tan^2\theta\,d\theta = \int (\sec^2\theta - 1)\,d\theta$$

$$= \tan\theta - \theta + C$$

Con $\tan\theta = \frac{x}{\sqrt{a^2-x^2}}$ y $\theta = \operatorname{arcsen}(x/a)$:

$$\int \frac{x^2\,dx}{(a^2-x^2)^{3/2}} = \frac{x}{\sqrt{a^2-x^2}} - \operatorname{arcsen}\frac{x}{a} + C$$

3. Sustitución Universal

en todas $z = \tan\frac{x}{2}$, $\operatorname{Sen} x = \frac{2z}{1+z^2}$, $\cos x = \frac{1-z^2}{1+z^2}$,

$dx = \frac{2\,dz}{1+z^2}$

4) $\int \frac{dx}{1-2\operatorname{sen}x}$ $\quad 1 - 2\operatorname{Sen} x = 1 - \frac{4z}{1+z^2} = \frac{z^2-4z+1}{1+z^2}$

$$\Rightarrow \int \frac{1+z^2}{z^2-4z+1} \cdot \frac{2\,dz}{1+z^2} = \int \frac{2\,dz}{z^2-4z+1}$$

Completando cuadrado: $z^2+4z+1=(z-2)^2-3$

$=(z-2)^2-(\sqrt{3})^2$. usando $\int \frac{du}{u^2-k^2}=\frac{1}{2k}\ln\left|\frac{u-k}{u+k}\right|$

$$2\,\frac{1}{2\sqrt{3}}\ln\left|\frac{z-2-\sqrt{3}}{z-2+\sqrt{3}}\right|+C=\frac{\sqrt{3}}{3}\ln\left|\frac{z-2-\sqrt{3}}{z-2+\sqrt{3}}\right|+C$$

$$\int\frac{dx}{1-2\cos x}=\frac{\sqrt{3}}{3}\ln\left|\frac{\tan(x/2)-2-\sqrt{3}}{\tan(x/2)-2+\sqrt{3}}\right|+C$$

b) $\int\frac{dx}{2-\cos x}$

$$2-\frac{1-z^2}{1+z^2}=\frac{2+2z^2-1+z^2}{1+z^2}=\frac{1+3z^2}{1+z^2}$$

$$\Rightarrow\int\frac{2\,dz}{1+3z^2}=\int\frac{2\,dz}{1+(\sqrt{3}z)^2}$$

Con $w=\sqrt{3}z$, $dz=dw/\sqrt{3}$: $\frac{2}{\sqrt{3}}\int\frac{dw}{1+w^2}=$

$$=\frac{2}{\sqrt{3}}\arctan w+C$$

$$\int\frac{dx}{2-\cos x}=\frac{2}{\sqrt{3}}\arctan\left(\sqrt{3}\tan\frac{x}{2}\right)+C$$

Michael Naranjo

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3) $\int \frac{dx}{3+5\sin x}$

$3+\frac{10z}{1+z^2} = \frac{3z^2+10z+3}{1+z^2} = \frac{(3z+1)(z+3)}{1+z^2}$

$\Rightarrow \int \frac{2\,dz}{(3z+1)(z+3)}$

Fracciones Parciales $\frac{2}{(3z+1)(z+3)} = \frac{A}{3z+1} + \frac{B}{z+3}$,

$2 = A(z+3) + B(3z+1)$

$z=-3 : 2 = -8B \Rightarrow B = -\frac{1}{4}$ $\quad z=-\frac{1}{3} : 2 = \frac{8}{3}A \Rightarrow A = \frac{3}{4}$

$\int \left(\frac{3/4}{3z+1} - \frac{1/4}{z+3}\right) dz = \frac{1}{4}\ln|3z+1| - \frac{1}{4}\ln|z+3|$

$+C$

$\int \frac{dx}{3+5\sin x} = \frac{1}{4}\ln\left|\frac{3\tan(x/2)+1}{\tan(x/2)+3}\right| + C$

$\int \frac{dx}{5+3\sin x}$

$5+\frac{6z}{1+z^2} = \frac{5z^2+6z+5}{1+z^2} \Rightarrow \int \frac{2\,dz}{5z^2+6z+5} =$

$\frac{2}{5}\int \frac{dz}{z^2+\frac{6}{5}z+1}$

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Completando Cuadrado

$$z^2 + \frac{6}{5}z + 1 = \left(z + \frac{3}{5}\right)^2 + \frac{16}{25} \cdot \text{con} \int \frac{du}{u^2 + k^2} = \frac{1}{k}$$

$$\arctan \frac{u}{k}, \quad k = \frac{4}{5}$$

$$\frac{2}{5} \cdot \frac{5}{4} \arctan \frac{z + 3/5}{4/5} + C = \frac{1}{2} \arctan \frac{5z+3}{4} + C$$

$$\int \frac{dx}{5 + 3\,\text{Sen}\,x} = \frac{1}{2} \arctan \frac{3 + 5\tan(x/2)}{4} + C$$

5) $$\int \frac{dx}{\text{Sen}\,x + \cos x - 1}$$

$$\frac{2z}{1+z^2} - \frac{1-z^2}{1+z^2} - \frac{1+z^2}{1+z^2} = \frac{2z - 1 + z^2 - 1 - z^2}{1+z^2}$$

$$= \frac{2z-2}{1+z^2}$$

$$\int \frac{1+z^2}{2(z-1)} \cdot \frac{2\,dz}{1+z^2} = \int \frac{dz}{z-1} = \ln|z-1| + C$$

$$\int \frac{dx}{\text{Sen}\,x - \cos x - 1} = \ln\left|\tan\frac{x}{2} - 1\right| + C$$

6) $\int \frac{5\operatorname{sen} x}{1+5\operatorname{sen}^2 x}\,dx$

$1+5\operatorname{sen}^2 x = 1 + \frac{4z^2}{(1+z^2)^2} = \frac{(1+z^2)^2+4z^2}{(1+z^2)^2}$

$= \frac{z^4+6z^2+1}{(1+z^2)^2}$

Sea $w=z^2$, $dw = 2z\,dz$: $\int \frac{2\,dw}{w^2+6w+1} = \int \frac{2\,dw}{(w+3)^2-8}$

con $K = \sqrt{8} = 2\sqrt{2}$

$$2\cdot\frac{1}{2\cdot 2\sqrt{2}} \ln\left|\frac{w+3-2\sqrt{2}}{w+3+2\sqrt{2}}\right| + C = \frac{\sqrt{2}}{4}\ln$$

$$\left|\frac{w+3-2\sqrt{2}}{w+3+2\sqrt{2}}\right| + C$$

$$\int \frac{5\operatorname{sen} x\,dx}{1+5\operatorname{sen}^2 x} = \frac{\sqrt{2}}{4}\ln\left|\frac{\tan^2(x/2)+3-2\sqrt{2}}{\tan^2(x/2)+3+2\sqrt{2}}\right| + C$$

4) Areas de regiones Planas

2. $x=\sqrt{y}$, $x=2-y$, eje x

$\sqrt{y} = 2-y \Rightarrow y = (2-y)^2 \Rightarrow y^2-5y+4=0 \Rightarrow x=1$ ($y=4$ no sirve pues $2-4<0$) Punto $(1,1)$.

Integramos Respecto a Y, de $y=0$ (eje x) a $y=1$. Derecha:

$x = 2-y$; Izquierda: $x=\sqrt{y}$

$$A = \int_0^1 \left[(2 - y) - \sqrt{y} \right] dy = \left[2y - \frac{y^2}{2} + \frac{2}{3} y^{3/2} \right]_0^1 = 2 - \frac{1}{2} - \frac{2}{3} =$$

$$= \frac{12 - 3 - 4}{6}$$

$$A = \frac{5}{6} u^2$$

4) $x = y^2$, $x = 2 - y^2$

Intersección $y^2 = 2 - y^2 \Rightarrow y^2 = 1 \Rightarrow y = \pm 1$; Puntos $(1, \pm 1)$

Derecha: $x = 2 - y^2$; Izquierda: $x = y^2$

$$A = \int_{-1}^{1} \left[(2 - y^2) - y^2 \right] dy = \int_{-1}^{1} (2 - 2y^2) dy = 2 \int_0^1 (2 - 2y^2) dy = 2$$

$$\left[2y - \frac{2y^3}{3} \right]_0^1 = 2 \left(2 - \frac{2}{3} \right) \qquad A = \frac{8}{3} u^2$$

6) $y = x^2$, $y = 2(x^2 + 1)^{-1}$

Intersección $x^2 = \frac{2}{x^2 + 1} \Rightarrow x^4 + x^2 - 2 = 0 \Rightarrow (x^2 + 2)(x^2 - 1)$

$= 0 \Rightarrow x = \pm 1$

en $x = 0$: $\frac{2}{1} = 2 > 0$ luego la curva superior es $y = \frac{2}{x^2 + 1}$

Por simetría (Funciones Pares)

$$A = \int_{-1}^{1} \left(\frac{2}{x^2 + 1} - x^2 \right) dx = 2 \left[2 \arctan x - \frac{x^3}{3} \right]_0^1 = 2 \left(2 \cdot \frac{\pi}{4} - \frac{1}{3} \right)$$

$$A = \pi - \frac{2}{3} \simeq 2.476 \, u^2$$

ESTILO

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9) $y = \frac{x}{x^2+1}$, $x=0$, $x=3$ (y el eje x)

En $[0,3]$ la función es ≥ 0. Sustitución $u = x^2+1$, $du = 2x\,dx$ ($x=0 \to u=1$, $x=3 \to u=10$).

$$A = \int_0^3 \frac{x}{x^2+1}\,dx = \frac{1}{2}\int_1^{10} \frac{du}{u} = \frac{1}{2}\ln u\Big|_1^{10} = \frac{1}{2}(\ln 10 - \ln 1)$$

$$A = \frac{1}{2}\ln 10 \approx 1.151\ u^2$$

10) Círculo de Radio

Circunferencia $x^2+y^2=R^2$, semicírculo superior $y = \sqrt{R^2-x^2}$. Por simetría, $A = 4\int_0^R \sqrt{R^2-x^2}\,dx$.

Sustitución $x = R\operatorname{Sen}\theta$, $dx = R\cos\theta\,d\theta$; $x=0 \to \theta=0$, $x=R \to \theta = \frac{\pi}{2}$

$$A = 4\int_0^{\pi/2} R\cos\theta \cdot R\cos\theta\,d\theta = 4R^2\int_0^{\pi/2} \frac{1+\cos 2\theta}{2} = 2R^2$$

$$\left[\theta + \frac{\operatorname{Sen} 2\theta}{2}\right]_0^{\pi/2} = 2R^2 \cdot \frac{\pi}{2} \qquad A = \pi R^2$$

5) Volumen de sólidos con sección conocida

Fórmula: $V = \int_a^b A(x)\,dx$, donde $A(x)$ es el área de la sección perpendicular al eje x

7. Base Circular de radio 1; Secciones Perpendiculares triangulos equilateros con un lado en la base

Base: $x^2 + y^2 \leq 1$, en cada $x \in [-1,1]$ el lado del triangulo es la cuerda: $\ell(x) = 2\sqrt{1-x^2}$

area de un triangulo equilatero: $A = \frac{\sqrt{3}}{4}\ell^2$

$\Rightarrow A(x) = \frac{\sqrt{3}}{4} \cdot 4(1-x^2) = \sqrt{3}(1-x^2)$

$$V = \int_{-1}^{1} \sqrt{3}(1-x^2)\,dx = 2\sqrt{3}\left[x - \frac{x^3}{3}\right]_0^1 = 2\sqrt{3} \cdot \frac{2}{3}$$

$$V = \frac{4\sqrt{3}}{3} \approx 2.309\,u^3$$

2. Base en el primer cuadrante bajo $y = 1 - \frac{x^2}{4}$, eje X y eje Y; Secciones: Cuadrados.

Corte con el eje X: $1 - \frac{x^2}{4} = 0 \Rightarrow x = 2$. En cada $x \in [0,2]$, el lado del cuadrado es $\ell(x) = 1 - \frac{x^2}{4}$

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Entonces $A(x) = \left(1 - \frac{x^2}{4}\right)^2 = 1 - \frac{x^2}{2} + \frac{x^4}{16}$

$$V = \int_0^1 \left(1 - \frac{x^2}{2} + \frac{x^4}{16}\right) dx = \left[x - \frac{x^3}{6} + \frac{x^5}{80}\right]_0^2 = 2 - \frac{8}{6} + \frac{32}{80} = 2$$

$$-\frac{4}{3} + \frac{2}{5} = \frac{30 - 20 + 6}{15}$$

$$V = \frac{16}{15} \approx 1.0673$$